

Virtual Humanoid User Flow

Aarav — Talent Nurturing System Desktop (Aarav-TN)

AI Execution Specification for the Simulated macOS-Style Web Application

Version 2.1 • Created 2026-04-29 • Built by Centific AI

Section 1 — Prototype Overview

Aarav is a virtual humanoid agent that operates a simulated macOS-style desktop web application — the Talent Nurturing System v2 (TN-v2). Aarav walks reviewers through training assignments, HIL approval queues, escalations, certifications, and analytics — all backed by a live PostgreSQL database via a FastAPI backend. This document is the authoritative AI execution specification. It defines every screen, interactive element, action record, narration pattern, and error-recovery rule that Aarav must follow when executing a Playwright (or equivalent) automated script.

Field	Value
Prototype Application	Aarav — Talent Nurturing System Desktop (Aarav-TN)
Primary User Role	Talent Lead / HR Reviewer watching Aarav demo the platform
Execution Actor	Virtual humanoid browser agent
Main Desktop URL	http://134.33.132.134/aarav-aegis-fe
Viewport Baseline	1440 × 900, 100% zoom
Coordinate Origin	Top-left of #desktop element
Primary Business Flow	Open Analytics → inspect KPIs → open Tracker → review assignments → approve HIL items → resolve escalations → view certs → audit log
Primary Audience	Reviewer watching Aarav navigate and explain each module
Created / Updated	2026-04-29
Template Status	Filled from Aarav-TN v2.1 frontend source code

Section 2 — Complete Flow Identification

Field	Value
flow_id	FLOW-AARAV-TN-FULL-DEMO
flow_name	Aarav Full Platform Walkthrough
flow_version	2.1
business_goal	Walk a reviewer through all six screens of the Talent Nurturing System using live DB data

Field	Value
persona	Senior Talent Lead reviewing Aarav's AI-assisted recommendations
trigger_utterance	"Open the Talent Nurturing System and walk me through the platform."
entry_screen_id	SCR-001-DESKTOP
expected_end_screen_id	SCR-007-AUDIT (ends on Audit Log)
narration_mode	Detailed guided walkthrough
evidence_required	Yes — capture screenshot on each major screen change

Section 3 — Required Runtime Parameters

Parameter	Req'd	Type	How Aarav Uses It
api_base_url	Yes	URL	Base for all /api/* fetch calls via Vite proxy
session_id	Yes	String	Correlates all actions in one transcript
user_role	Yes	String	"Talent Lead" — drives narration depth
current_screen	Yes	Enum	Tracks which dock icon was clicked last
dark_mode	No	Boolean	Pass true to activate dark mode via #dark-mode-toggle-btn
auto_refresh_seconds	No	Number	DataContext polls every 10s — default interval
window_open_settle_ms	Yes	Number	Wait 310ms after dock click before targeting inner elements
viewport_width	Yes	Number	1440
viewport_height	Yes	Number	900
speaker_style	Yes	Enum	"Friendly, technical, data-aware"

Section 4 — UI/UX Screen Inventory

Screen ID	Name	Route	Purpose	Expected Evidence
SCR-001-DESKTOP	Desktop Home	/	Wallpaper with clock, widgets, dock	#desktop renders; dock visible; clock time shows
SCR-002-ANALYTICS	Analytics	window	KPI cards + donut chart + bar chart	8 KPI cards render; donut slices visible
SCR-003-TRACKER	Assignment Tracker	window	Full assignment table with filter	Table rows render; filter buttons active
SCR-004-ESCALATIONS	Escalation Panel	window	Cards for open escalations with action buttons	Escalation cards visible; Extend/Replace/Accept buttons present
SCR-005-HIL	HIL Approval Queue	window	Table of pending HIL items with Approve/Reject	HIL rows render; approve/reject btns present

Screen ID	Name	Route	Purpose	Expected Evidence
SCR-006-CERTS	Certification Registry	window	Cert table with filter; click for journey modal	Cert rows render; filter buttons present
SCR-007-AUDIT	Audit Log	window	Append-only audit event table	#audit-log-table rows render
SCR-008-AARAV-ORB	Aarav AI Panel	overlay	Floating AI insights from live data	#aarav-panel renders; insight cards visible
SCR-009-MODAL	Modal Overlay	overlay	Detail drilldown for any row or action	Modal content renders; close button present

Section 5 — Screen Layout Region Map

Coordinates assume 1440 × 900. All regions sourced from index.css and component inspection.

Region ID	Applies To	Pixel Bounds	Description	Aarav Instruction
REG-DESKTOP-ROOT	All	0,0,1440,900	#desktop full viewport	Confirm page loaded via #desktop
REG-DOCK	All	center-bottom ~92px	#dock-container — glassmorphic floating dock	All dock item clicks happen here
REG-WINDOW	When open	inset ~8px all sides	.window element — screen host	All screen interactions inside this region
REG-SCREEN-HEADER	Window	0,0>window-width,48	.screen-header — title bar with time/API status/controls	Read API status dot; click maximize/close buttons
REG-SCREEN-BODY	Window	0,48,~1100,900	.screens-wrap — screen content area	All screen content renders here
REG-LIVE-PANEL	Window	~1100,0,1440,900	LivePanel — right-side audit feed	Toggle via #livepanel-toggle-btn
REG-CLOCK	Desktop	center, top 50%	Large clock — only visible when no window is open	Read time for timestamp narration
REG-WIDGETS	Desktop	top-right, ~230px wide	.desk-widget — Agent Notes + Today's Tasks	Read task count and notes
REG-AARAV-ORB	All	bottom-right corner	#aarav-orb-trigger — floating AI orb button	Click to open AI insights panel

Section 6 — Element Target Registry

All interactive elements in the application with their exact DOM IDs, CSS selectors, and required actions. Element IDs are derived directly from the frontend JSX source code.

6A — Dock Navigation Buttons

Elem ID	Screen	DOM ID	Selector / Locator	Required Action	Expected Result
EL-001	SCR-001	dock-home-btn	#dock-home-btn	Click	goHome() — window closes, data refreshes
EL-002	SCR-001	dock-analytics-btn	#dock-analytics-btn	Click	Analytics screen opens
EL-003	SCR-001	dock-tracker-btn	#dock-tracker-btn	Click	Tracker screen opens
EL-004	SCR-001	dock-escalations-btn	#dock-escalations-btn	Click	Escalations screen opens
EL-005	SCR-001	dock-hil-btn	#dock-hil-btn	Click	HIL queue screen opens
EL-006	SCR-001	dock-certs-btn	#dock-certs-btn	Click	Certifications screen opens
EL-007	SCR-001	dock-audit-btn	#dock-audit-btn	Click	Audit log screen opens

6B — Window Chrome Controls

Elem ID	Screen	DOM ID	Selector	Required Action	Expected Result
EL-010	Any window	window-close-btn	#window-close-btn	Click	Window closes; home state
EL-011	Any window	window-maximize-btn	#window-maximize-btn	Click	Window maximizes/restores
EL-012	Any window	dark-mode-toggle-btn	#dark-mode-toggle-btn	Click	Dark/light mode toggles
EL-013	Any window	livepanel-toggle-btn	#livepanel-toggle-btn	Click	Live panel expands/collapses

6C — Analytics Screen Elements

Elem ID	DOM ID / Pattern	Selector	Required Action	Expected Result
EL-020	kpi-{key}-card	#kpi-assign-card, #kpi-comp-card, etc.	Click	KPI detail modal opens
EL-021	kpi-detail-close-btn	#kpi-detail-close-btn	Click	Modal closes
EL-022	kpi-detail-dismiss-btn	#kpi-detail-dismiss-btn	Click	Modal closes
EL-023	status-{status}-slice	#status-complete-slice, #status-active-slice, etc.	Click	Status detail modal opens
EL-024	status-detail-close-btn	#status-detail-close-btn	Click	Modal closes

Elem ID	DOM ID / Pattern	Selector	Required Action	Expected Result
EL-025	rfp-bar-{ref}-bar	#rfp-bar-0-bar (indexed by position)	Click	RFP detail modal opens
EL-026	rfp-detail-close-btn	#rfp-detail-close-btn	Click	Modal closes

6D — Tracker Screen Elements

Elem ID	DOM ID / Pattern	Selector	Required Action	Expected Result
EL-030	tracker-filter-all-btn	#tracker-filter-all-btn	Click	Filter resets to all assignments
EL-031	tracker-filter-{status}-btn	#tracker-filter-active-btn, etc.	Click	Table filters to that status
EL-032	tracker-row-{id}	#tracker-row-{assignment_id}	Click	Assignment detail modal opens
EL-033	assignment-detail-close-btn	#assignment-detail-close-btn	Click	Assignment modal closes

6E — HIL Approval Screen Elements

Elem ID	DOM ID / Pattern	Selector	Required Action	Expected Result
EL-040	hil-approve-{id}-btn	#hil-approve-{hil_id}-btn	Click	HIL confirm approval modal opens
EL-041	hil-reject-{id}-btn	#hil-reject-{hil_id}-btn	Click	HIL confirm rejection modal opens
EL-042	hil-confirm-action-btn	#hil-confirm-action-btn	Click	POST to /api/hil-queue/{id}/action; row removed
EL-043	hil-confirm-cancel-btn	#hil-confirm-cancel-btn	Click	Modal closes; no action
EL-044	hil-confirm-close-btn	#hil-confirm-close-btn	Click	Modal closes; no action
EL-045	hil-act-notes	#hil-act-notes	Type text	Notes textarea accepts input

6F — Escalations Screen Elements

Elem ID	DOM ID / Pattern	Selector	Required Action	Expected Result
EL-050	escalation-extend-{id}-btn	#escalation-extend-{esc_id}-btn	Click	Extend Deadline modal opens
EL-051	escalation-replace-{id}-btn	#escalation-replace-{esc_id}-btn	Click	Replace Resource confirm modal opens
EL-052	escalation-accept-{id}-btn	#escalation-accept-{esc_id}-btn	Click	Accept Risk confirm modal opens

Elem ID	DOM ID / Pattern	Selector	Required Action	Expected Result
EL-053	extend-deadline-confirm-btn	#extend-deadline-confirm-btn	Click	POST resolveEscalation with extend action
EL-054	extend-deadline-cancel-btn	#extend-deadline-cancel-btn	Click	Modal closes; no action
EL-055	extend-deadline-close-btn	#extend-deadline-close-btn	Click	Modal closes; no action
EL-056	nd-inp	#nd-inp	Fill date value	Date input accepts new deadline
EL-057	nd-notes	#nd-notes	Type text	Reason textarea accepts input
EL-058	escalation-confirm-resolve-btn	#escalation-confirm-resolve-btn	Click	Escalation resolved; card removed
EL-059	escalation-confirm-cancel-btn	#escalation-confirm-cancel-btn	Click	Modal closes; no action
EL-060	escalation-confirm-close-btn	#escalation-confirm-close-btn	Click	Modal closes; no action
res-notes	res-notes	#res-notes	Type text	Notes textarea accepts input

6G — Certifications Screen Elements

Elem ID	DOM ID / Pattern	Selector	Required Action	Expected Result
EL-070	certs-filter-all-btn	#certs-filter-all-btn	Click	Shows all certifications
EL-071	certs-filter-registered-btn	#certs-filter-registered-btn	Click	Filters to registered certs
EL-072	certs-filter-verified-btn	#certs-filter-verified-btn	Click	Filters to verified certs
EL-073	certs-filter-pending-btn	#certs-filter-pending-btn	Click	Filters to pending certs
EL-074	cert-row-{id}	#cert-row-{cert_id}	Click	Certification journey modal opens
EL-075	cert-journey-close-btn	#cert-journey-close-btn	Click	Journey modal closes

6H — Audit Log Screen Elements

Elem ID	DOM ID / Pattern	Selector	Required Action	Expected Result
EL-080	audit-log-table	#audit-log-table	Read rows	Audit entries visible
EL-081	audit-row-{id}	#audit-row-{log_id}	Read only	Row data: time, level, RFP, action

6I — Aarav AI Orb Elements

Elem ID	DOM ID / Pattern	Selector	Required Action	Expected Result
EL-090	aarav-orb-trigger	#aarav-orb-trigger	Click	AI insights panel opens
EL-091	aarav-panel	#aarav-panel	Read content	AI insight cards visible with live data
EL-092	aarav-panel-close-btn	#aarav-panel-close-btn	Click	AI panel closes
EL-093	aarav-view-{idx}-btn	#aarav-view-0-btn, etc.	Click	View details for that insight
EL-094	aarav-action-{idx}-btn	#aarav-action-0-btn, etc.	Click	Take action on that insight
EL-095	aarav-backdrop	#aarav-backdrop	Click	AI panel closes

6J — Live Panel Elements

Elem ID	DOM ID	Selector	Required Action	Expected Result
EL-100	livepanel-toggle-btn	#livepanel-toggle-btn	Click	Live panel expands/collapses
EL-101	livepanel-collapsed-label	#livepanel-collapsed-label	Click	Collapsed panel expands

Section 7 — Standard AI Action Record Template

Each action follows this JSON schema. The runtime iterates actions[] and executes top-to-bottom. All element IDs must match Section 6 exactly.

7.1 Schema (Canonical Shape)

```
{  "action_id": "ACT-### (matches Section 8)",  "screen_id": "SCR-### (from Section 4)",  "action_type": "click | type | fill_date | read_text | screenshot | validate | wait_for_element | recover",  "target": {    "element_id": "EL-### (must exist in Section 6)",    "selector": "#exact-dom-id",    "fallback_region": "REG-... (from Section 5)",    "fallback_coordinates": {      "x": 0, "y": 0    }  },  "input_parameters": {    "any runtime param": "value"  },  "preconditions": ["human-readable conditions"],  "execution_steps": ["ordered browser ops"],  "expected_response": {    "screen_after": "SCR-###",    "dom_changes": ["selectors that appear/activate"],    "network_calls": [{      "method": "GET|POST",      "url_template": "/api/...",      "expected_status": 200    }],    "state_after": {      "currentScreen": "screenKey"    }  },  "validation_rules": ["predicate strings; all must pass"],  "humanoid_speaker_notes": {    "before_action": "what Aarav says before",    "during_action": "narration while waiting",    "after_action": "what Aarav says once validation passes"  },  "failure_recovery": [{    "if": "predicate",    "do": "action",    "max_retries": 1  }],  "timing": {    "wait_before_action_ms": 310,    "stable_for_ms": 100,    "max_wait_ms": 15000  } }
```

Section 8 — Detailed Action Matrix (All Action Types)

ACT	Type	Screen	Target EL	Parameters	Validation	Speaker Note	Fallback
ACT-001	open_url	SCR-001	—	http://localhost:5173	#desktop rendered	"The Talent Nurturing desktop is open."	Refresh tab
ACT-002	wait_render	SCR-001	—	—	#dock-container visible	"Waiting for dock to load."	Wait 2s; retry
ACT-003	read_text	SCR-001	—	—	Clock text HH:MM	"It is currently {time}."	Skip; continue
ACT-004	read_widget	SCR-001	—	—	.dw-notes and .dw-tasks render	Read task count aloud	Skip; continue
ACT-010	click	SCR-001	EL-002	#dock-analytics-btn	.window.open renders	"Opening Analytics dashboard."	Retry click after 1s
ACT-011	wait	SCR-001→2	—	wait_before: 310ms	.screen-header visible	"Analytics is loading."	Wait 500ms more
ACT-012	read_kpi	SCR-002	EL-020	#kpi-assign-card	Text contains %	"Assignment rate is {val}%."	Log; continue
ACT-013	click	SCR-002	EL-020	#kpi-assign-card	Modal appears	"Drilling into assignment rate."	Skip drilldown
ACT-014	click	SCR-002	EL-021	#kpi-detail-close-btn	Modal disappears	"Closing detail view."	Click #kpi-detail-dismiss-btn
ACT-015	read	SCR-002	EL-023	#status-complete-slice	Donut renders	"Donut shows status breakdown."	Skip; continue
ACT-016	click	SCR-002	EL-025	#rfp-bar-0-bar	RFP modal appears	"Clicking first RFP bar."	Try bar at index 1
ACT-017	click	SCR-002	EL-026	#rfp-detail-close-btn	Modal closes	"Closing RFP detail."	Click #rfp-detail-dismiss-btn
ACT-020	click	SCR-002	EL-003	#dock-tracker-btn	Tracker screen shows	"Switching to assignment tracker."	Retry after 1s
ACT-021	read	SCR-003	EL-032	#tracker-row-{any id}	Row count > 0	"{n} assignments loaded."	Check /api/assignments
ACT-022	click	SCR-003	EL-030	#tracker-filter-all-btn	Filter active=all	"Showing all assignments."	Log; proceed
ACT-023	click	SCR-003	EL-031	#tracker-filter-active-btn	Rows filter to active	"Filtered to active only."	Log; proceed
ACT-024	click	SCR-003	EL-032	#tracker-row-{first_id}	Detail modal opens	"Reviewing assignment details."	Skip modal
ACT-025	click	SCR-003	EL-033	#assignment-detail-close-btn	Modal closes	"Closing detail view."	Click outside modal

ACT	Type	Screen	Target EL	Parameters	Validation	Speaker Note	Fallback
ACT-030	click	SCR-003	EL-005	#dock-hil-btn	HIL screen shows	"Opening HIL approval queue."	Retry after 1s
ACT-031	read	SCR-005	EL-040	#hil-approve-{id}-btn	Button exists	"Pending HIL item found."	Skip if queue empty
ACT-032	click	SCR-005	EL-040	#hil-approve-{id}-btn	Confirm modal opens	"Initiating approval."	Use reject flow instead
ACT-033	type	SCR-005	EL-045	#hil-act-notes	Textarea accepts text	"Adding approval notes."	Skip; confirm without notes
ACT-034	click	SCR-005	EL-042	#hil-confirm-action-btn	Row removed from table	"HIL approved. Assignment active."	Abort; check backend
ACT-040	click	SCR-005	EL-004	#dock-escalations-btn	Escalations screen shows	"Opening escalation panel."	Retry after 1s
ACT-041	read	SCR-004	EL-050	#escalation-extend-{id}-btn	Button exists	"Open escalation found."	Skip if none open
ACT-042	click	SCR-004	EL-050	#escalation-extend-{id}-btn	Extend deadline modal opens	"Opening deadline extension."	Try accept risk instead
ACT-043	fill_date	SCR-004	EL-056	#nd-inp	Date field has value	"Setting new deadline date."	Show toast warning
ACT-044	click	SCR-004	EL-053	#extend-deadline-confirm-btn	Escalation resolved	"Deadline extended successfully."	Abort; check backend
ACT-050	click	SCR-004	EL-006	#dock-certs-btn	Certs screen shows	"Opening certification registry."	Retry after 1s
ACT-051	click	SCR-006	EL-070	#certs-filter-all-btn	All certs shown	"Showing all certifications."	Log; proceed
ACT-052	click	SCR-006	EL-074	#cert-row-{first_id}	Journey modal opens	"Reviewing cert journey."	Skip modal
ACT-053	click	SCR-006	EL-075	#cert-journey-close-btn	Modal closes	"Journey reviewed."	Click outside modal
ACT-060	click	SCR-006	EL-007	#dock-audit-btn	Audit screen shows	"Opening audit log."	Retry after 1s
ACT-061	read	SCR-007	EL-080	#audit-log-table	Row count > 0	"Audit log shows {n} events."	Check /api/audit-logs
ACT-070	click	SCR-007	EL-090	#aarav-orb-trigger	AI panel opens	"Opening Aarav AI insights."	Retry after 1s
ACT-071	read	SCR-008	EL-091	#aarav-panel	Insight cards visible	"AI has generated {n} insights."	Log empty; continue

ACT	Type	Screen	Target EL	Parameters	Validation	Speaker Note	Fallback
ACT-072	click	SCR-008	EL-092	#aarav-panel-close-btn	Panel closes	"Closing AI insights."	Click #aarav-backdrop
ACT-080	click	Any	EL-001	#dock-home-btn	Window closes; data refreshes	"Returning home; refreshing data."	Click #window-close-btn
ACT-090	screenshot	Any	—	tag: screen-change	File saved	"Capturing screenshot."	Log; continue
ACT-099	final_summary	Any	—	summary_style	Response generated	"Full platform walkthrough complete."	—

Section 9 — Complete Example Flow: Aarav Full Platform Demo

Seq	ACTs	Screen	Aarav Does	UI Target	Expected Response	Speaker Notes
1	001–003	SCR-001-DESKTOP	Open app; read clock + widgets	#desktop, clock	Wallpaper, dock, widgets visible	"The TN desktop is live."
2	010–017	SCR-002-ANALYTICS	Click Analytics dock icon; read all 8 KPIs; drilldown one KPI; inspect donut; click bar chart	EL-002, EL-020–EL-026	KPI cards render; modal opens/closes; bar chart clickable	"Assignment rate is {val}%. Drilling in..."
3	020–025	SCR-003-TRACKER	Click Tracker; filter to active; click first row; review progress circle	EL-003, EL-030–EL-033	Table renders; modal shows circular progress	"Active assignment: {name}. Progress: {p}%."
4	030–034	SCR-005-HIL	Click HIL dock; read pending items; approve first item with notes	EL-005, EL-040–EL-045	Row removed post-approval; toast shows	"HIL approved. Assignment is now active."
5	040–044	SCR-004-ESCALATIONS	Click Escalations dock; extend deadline on first open escalation	EL-004, EL-050–EL-056	Extend modal opens; date submitted; card removed	"Deadline extended. Escalation resolved."
6	050–053	SCR-006-CERTS	Click Certs dock; show all; click first	EL-006, EL-070–EL-075	Cert table; journey modal with 8 stages	"Cert journey shows all 8 stages."

Seq	ACTs	Screen	Aarav Does	UI Target	Expected Response	Speaker Notes
			cert row for journey modal			
7	060–061	SCR-007-AUDIT	Click Audit dock; read table	EL-007, EL-080	Audit entries visible with time, level, action	"Audit log shows {n} events."
8	070–072	SCR-008-AARAV-ORB	Click Aarav orb; read AI insights; close panel	EL-090–EL-092	Insight cards from live data	"Aarav AI generated {n} insights."
9	080, 099	SCR-001-DESKTOP	Click home; final spoken summary	EL-001	Window closes; data refreshes	"Full demo complete. All modules reviewed."

Section 10 — Detailed Page Examples

10.1 Desktop Home (SCR-001-DESKTOP)

Field	Value
screen_id	SCR-001-DESKTOP
purpose	Starting screen. Aarav uses the dock to open all six app modules.
layout	Center: large clock (visible when no window open). Top-right: Agent Notes + Today's Tasks widgets. Bottom-center: floating glassmorphic dock with 7 icons + separators.
primary targets	#dock-home-btn, #dock-analytics-btn, #dock-tracker-btn, #dock-escalations-btn, #dock-hil-btn, #dock-certs-btn, #dock-audit-btn
magnification behavior	Icons grow from 54px to 82px when cursor nears them. When window is open, dock items shrink. When maximized, dock hides until cursor nears bottom edge.
dock_peeking trigger	isMaximized && clientY >= window.innerHeight - 60 → dockPeeking=true → bottom: 10px else bottom: -100px
AI observation	Confirm desktop loaded by detecting #dock-container, .dw-notes, .dw-tasks
narration_before_action	"The Talent Nurturing desktop is open. I can see the clock, the task widgets, and the dock."
success_validation	Within 3s the clicked module's .window.open class is present
fallback	If dock click does nothing, retry after 1s; if still inert, refresh page (DataContext polls every 10s)

10.2 Analytics Dashboard (SCR-002-ANALYTICS)

Field	Value
entry	Click #dock-analytics-btn → wait 310ms → screen-header appears

Field	Value
KPI cards	8 cards: assign, comp, pass, over, cert, hil, cap, time. Each has DOM id #kpi-{key}-card. Click → detail modal.
donut chart	Status breakdown. Each slice has id #status-{status}-slice (e.g. #status-complete-slice). Click → detail modal.
bar chart	RFP compliance. Each bar has id #rfp-bar-{i}-bar. Click → RFP detail modal.
alert badges	If overdue_count > 0 or pending_hil > 0 — system alerts shown at bottom of screen
validation	#kpi-assign-card text contains '%'; donut SVG renders; bar chart SVG renders
speaker_notes_before	"Opening the analytics dashboard to review key performance indicators."
speaker_notes_after	"All 8 KPI cards are loaded. Assignment rate is {val}%, completion rate is {val}%."

10.3 Assignment Tracker (SCR-003-TRACKER)

Field	Value
entry	Click #dock-tracker-btn → wait 310ms
filter buttons	#tracker-filter-all-btn, #tracker-filter-active-btn, #tracker-filter-at_risk-btn, #tracker-filter-overdue-btn, #tracker-filter-complete-btn, #tracker-filter-pending-btn
table rows	Each row has id #tracker-row-{assignment_id}. Click → assignment detail modal with circular progress chart.
modal content	Circular progress SVG, resource code, role, RFP, program, deadline, content/test/case_study status, scores
close modal	#assignment-detail-close-btn
validation	Row count > 0; click row → modal shows progress % ≥ 0

10.4 HIL Approval Queue (SCR-005-HIL)

Field	Value
entry	Click #dock-hil-btn → wait 310ms
approve flow	1) Click #hil-approve-{id}-btn → confirm modal → 2) optionally type in #hil-act-notes → 3) click #hil-confirm-action-btn → POST /api/hil-queue/{id}/action → row removed + toast
reject flow	1) Click #hil-reject-{id}-btn → confirm modal → 2) type reason in #hil-act-notes → 3) click #hil-confirm-action-btn → row removed + toast
cancel modal	#hil-confirm-cancel-btn or #hil-confirm-close-btn
validation	After approval: row with that id disappears; toast shows; pushLog called with hil_approve action_type
empty state	If hilItems.length === 0, a no-items message displays. Do not attempt clicks on non-existent rows.

10.5 Escalation Panel (SCR-004-ESCALATIONS)

Field	Value
entry	Click #dock-escalations-btn → wait 310ms
Extend flow	1) #escalation-extend-{id}-btn → modal opens → 2) fill #nd-inp with date → 3) optionally type #nd-notes → 4) #extend-deadline-confirm-btn → POST resolveEscalation → card removed
Replace flow	1) #escalation-replace-{id}-btn → confirm modal → 2) optionally type #res-notes → 3) #escalation-confirm-resolve-btn → resolved
Accept flow	1) #escalation-accept-{id}-btn → confirm modal → 2) #escalation-confirm-resolve-btn → resolved
cancel	#extend-deadline-cancel-btn, #extend-deadline-close-btn, #escalation-confirm-cancel-btn, #escalation-confirm-close-btn
validation	Date #nd-inp must not be empty before confirm. Toast shows success. Card status changes.
guard	If nd value is empty, showToast('⚠ Please select a new deadline') fires — do not submit.

10.6 Certification Registry (SCR-006-CERTS)

Field	Value
entry	Click #dock-certs-btn → wait 310ms
filter	#certs-filter-all-btn, #certs-filter-registered-btn, #certs-filter-verified-btn, #certs-filter-pending-btn
journey modal	Click #cert-row-{id} → modal with 8 stages shown as timeline: Gap Identified → Certified & Registered
modal close	#cert-journey-close-btn
validation	Cert table renders; click row → modal shows cert_name and resource name; 8 stage items visible

10.7 Audit Log (SCR-007-AUDIT)

Field	Value
entry	Click #dock-audit-btn → wait 310ms
table	#audit-log-table — columns: Time, Level, RFP, Resource ID, Action, Actor, Message
rows	Each row id #audit-row-{log_id} — read-only; no clickable actions
levels	info → badge b-pp, warning → b-wn, error → b-er, action → b-ok
validation	#audit-log-table tbody tr count > 0; time column shows locale date
live panel	The same audit events appear in the LivePanel feed on the right. Toggle via #livepanel-toggle-btn.

10.8 Aarav AI Orb (SCR-008-AARAV-ORB)

Field	Value
entry	Click #aarav-orb-trigger (bottom-right floating button) from any screen
panel	#aarav-panel slides open. Shows AI-derived insights from live escalations, HIL items, completed assignments, and metrics.
insight types	risk (from open escalations), recommendation (from pending HIL), approval (from completed assignments)
card buttons	Each insight card has #aarav-view-{idx}-btn and #aarav-action-{idx}-btn
close	#aarav-panel-close-btn or click #aarav-backdrop
validation	#aarav-panel opacity = 1 after open; insight card text references real resource names from DB
placeholder guardrail	If no live data: insights array may be empty. Narrate: "No AI insights at this time — data may still be loading."

Section 11 — Validation Rules and Evidence Capture

Validation Type	Rule	Example	Evidence to Capture
Page validation	.window.open class present after dock click	Analytics screen after clicking #dock-analytics-btn	Screenshot + .screen-header text
Element validation	Target element exists AND is visible before click	#hil-approve-{id}-btn visible before click	Element ID, selector, bounding box
Network validation	GET /api/* returns 200 with non-empty array	GET /api/assignments returns list	Response status + length
Data validation	Table rows > 0 for all screens	#audit-log-table tbody tr count > 0	Row count log
Action validation	After HIL approve: row removed from table	#hil-approve-{id}-btn no longer present	Before/after screenshots
Modal validation	Modal content renders within 500ms of click	#assignment-detail-close-btn present in DOM	Modal screenshot
Narration validation	Speaker notes reference only visible data	Do not quote a resource name not on screen	Narration transcript
API health validation	GET /api/health returns {status: 'healthy'}	api-dot element is green (#16A34A)	API status text in screen-header

Section 12 — Error Handling Matrix

Err ID	Scenario	Detection	Recovery	Max Retries	Final Aarav Response if Unresolved
ERR-001	Backend offline	.api-dot is red; GET /api/health fails	Wait 30s; retry. If still down, tell user.	2	"The backend is offline. I cannot load live data."

Err ID	Scenario	Detection	Recovery	Max Retries	Final Aarav Response if Unresolved
ERR-002	Empty assignments	Table has 0 rows after load	Wait 10s for DataContext poll; try /api/assignments directly	1	"No assignments in database. Schema may need seed data."
ERR-003	Empty HIL queue	No #hil-approve-* btn buttons found	Narrate empty state; skip HIL flow	0	"HIL queue is empty — no pending approvals."
ERR-004	No open escalations	No escalation cards rendered	Skip escalation flow; narrate clear status	0	"No open escalations — all resources on track."
ERR-005	Modal doesn't open	No modal-head in DOM after click	Retry click after 500ms	2	"The modal did not open; skipping detail view."
ERR-006	Date input empty on confirm	Toast shows '⚠ Please select a new deadline'	Re-fill #nd-inp before retry	1	"Date selection required — filling current date + 30 days."
ERR-007	Dock click no-op	Window class unchanged after 3s	Retry once; then refresh page	2	"Dock unresponsive; refreshing page."
ERR-008	CORS / network error	fetch throws; console error	Surface to user; wait for next 10s poll	1	"Network error reaching backend API."
ERR-009	Aarav orb insights empty	#aarav-panel has no insight cards	Narrate empty state; close panel	0	"No AI insights — data may still be initializing."

Section 13 — Humanoid Narration Script Patterns

Situation	Template	Example
Before opening a screen	"I am going to open [screen] to [reason]."	"I am going to open the HIL queue to review pending approvals."
While API loading	"[Endpoint] is loading. Once ready I will look for [items]."	"The assignments API is loading. Once ready I will look for active items."
After screen opens	"[Screen] is visible. I can see [regions/items]."	"Analytics is visible. I can see all 8 KPI cards loaded from the database."
Reading a KPI	"The [metric name] is [value][unit] — [interpretation]."	"The completion rate is 78% — most assignments are on track."
Before an action	"I will now [action] to [reason]."	"I will now approve this HIL item to activate the training assignment."

Situation	Template	Example
After action success	"[Action] succeeded. [Effect observed]."	"Approval succeeded. The row has been removed from the queue."
When uncertain / placeholder	"I can see [known], but [content] is not available yet."	"I can see the cert row, but the journey data has not loaded yet."
Recovering from error	"The expected [item] is missing, so I will try [fallback]."	"The modal did not open; retrying the row click."
Final summary	"We reviewed [screens/features]. The takeaway is [insight]. Next step: [action]."	"We reviewed Analytics, Tracker, HIL, Escalations, Certs, and Audit. All modules functional. Recommend scheduling a data seed for demo readiness."

Section 14 — Decision Logic and Branching

ID	Condition	Aarav Decision	Action	Narration
DEC-001	HIL queue is empty	Skip HIL approval steps	Navigate to next screen	"HIL queue is empty — no approvals needed today."
DEC-002	No open escalations	Skip escalation resolution	Navigate to certs	"No open escalations — all resources are on track."
DEC-003	date input empty on extend	Do not submit; fill date first	ACT-043 with today+30d	"Date required before confirming extension."
DEC-004	API health check fails	Wait 30s; do not click API-dependent elements	Park on desktop	"Backend unreachable. Waiting for auto-refresh."
DEC-005	Modal content is placeholder text	Mark as placeholder; skip narration	Log; close modal	"This content is not yet available in the database."
DEC-006	No assignments in tracker	Narrate empty state; skip row clicks	Navigate forward	"Tracker is empty — database may need seed data."
DEC-007	Window stuck maximized	Click #window-maximize-btn to restore	ACT-011 again	"Restoring window to normal size."
DEC-008	Aarav AI panel has no insights	Narrate empty; close panel immediately	ACT-072	"No AI insights available yet."

Section 15 — Master JSON for AI Runtime

```
{
  "flow_id": "FLOW-AARAV-TN-FULL-DEMO",
  "prototype": "Aarav Talent Nurturing System Desktop",
  "entry screen id": "SCR-001-DESKTOP",
  "parameters": {
    "api base url": "http://localhost:8000",
    "session id": "AARAV-SESSION-001",
    "viewport width": 1440,
    "viewport height": 900,
    "window_open_settle_ms": 310,
    "auto_refresh_seconds": 10,
    "speaker_style": "Friendly, technical, data-aware"
  },
  "runtime rules": {
    "preferred locator order": ["#id", ".class+text", "fallback_coordinates"],
    "wait_timeout_seconds": 15,
    "screenshot_policy": "capture_on_screen_change_and_modal_open",
    "narration_policy": "speak_before_and_after_each_screen",
    "do_not_invent_placeholder_content": true,
    "do_not_click_if_element_missing": true
  },
  "screens": [
    "SCR-001-DESKTOP", "SCR-002-ANALYTICS", "SCR-003-TRACKER", "SCR-004-ESCALATIONS", "SCR-005-HIL", "SCR-006-CERTS", "SCR-007-AUDIT", "SCR-008-AARAV-ORB"
  ],
  "completion_criteria": [
    "Desktop home loaded and clock"
  ]
}
```



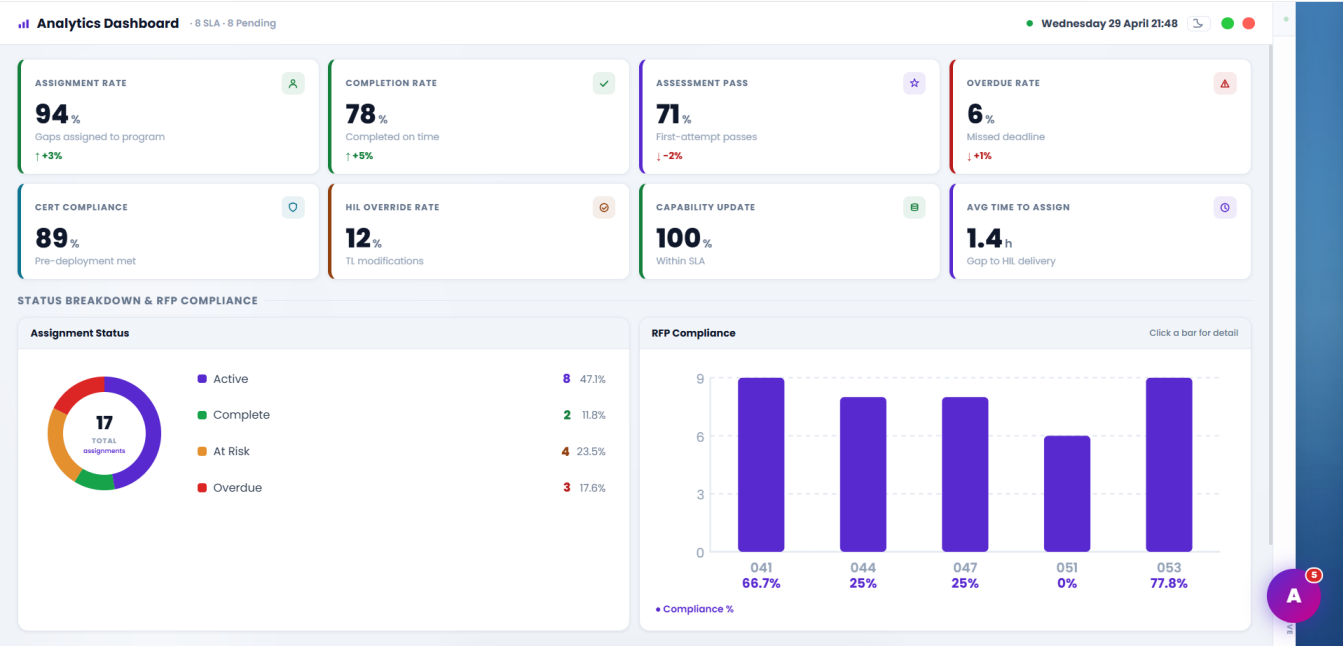
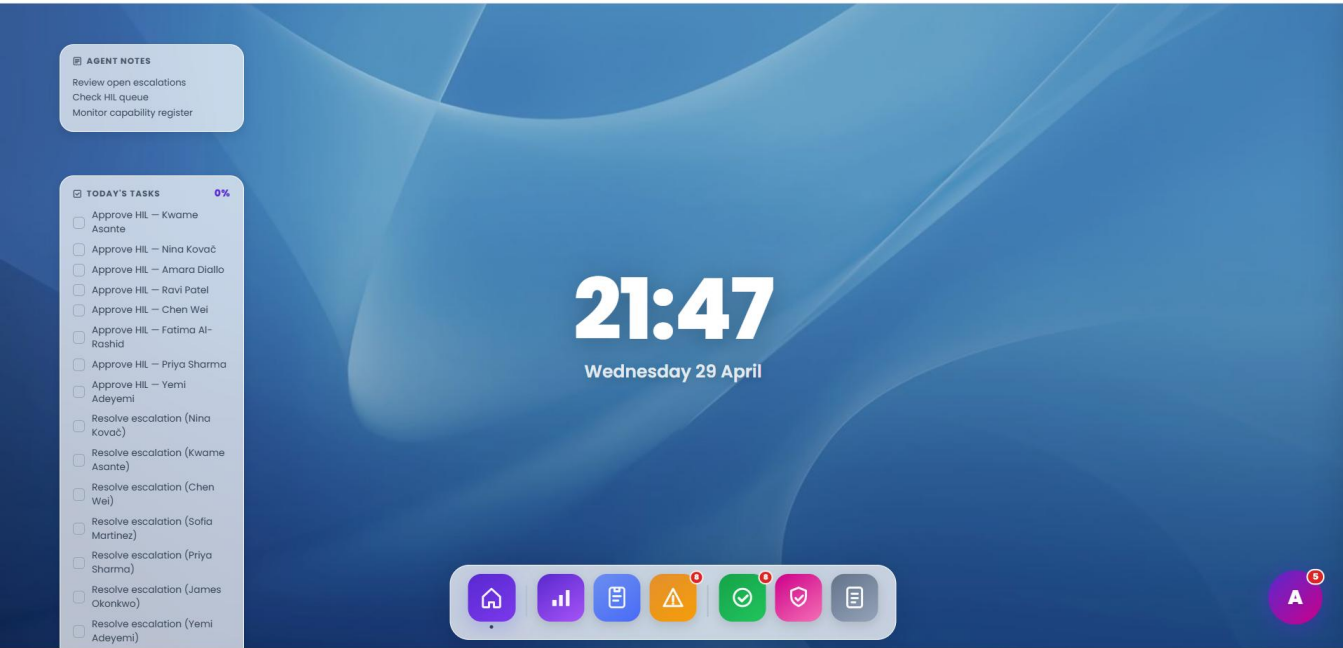
```
visible",      "All 8 KPI cards read from Analytics",      "At least one tracker row inspected",  
"HIL queue reviewed (approved or narrated empty)",      "Escalation reviewed (resolved or narrated  
clear)",      "Cert registry reviewed; journey modal opened",      "Audit log rows read",      "Aarav AI  
orb opened; insights narrated",      "Final summary delivered"  ] }
```

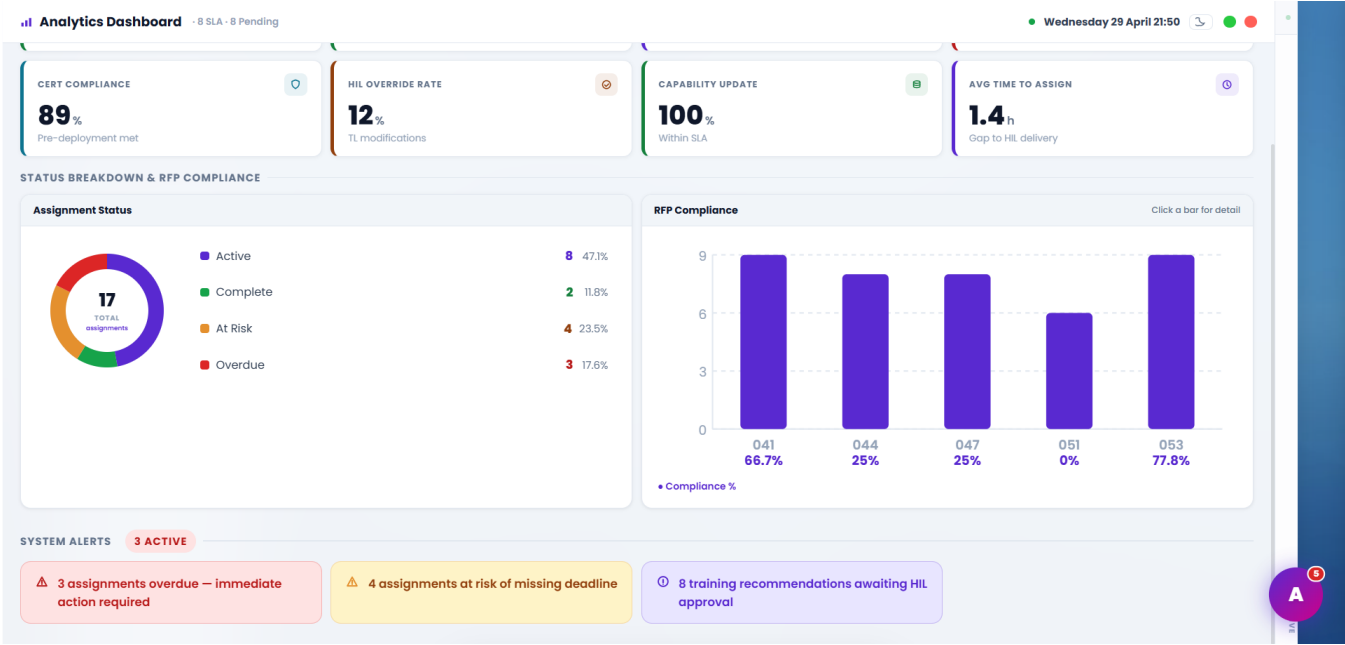
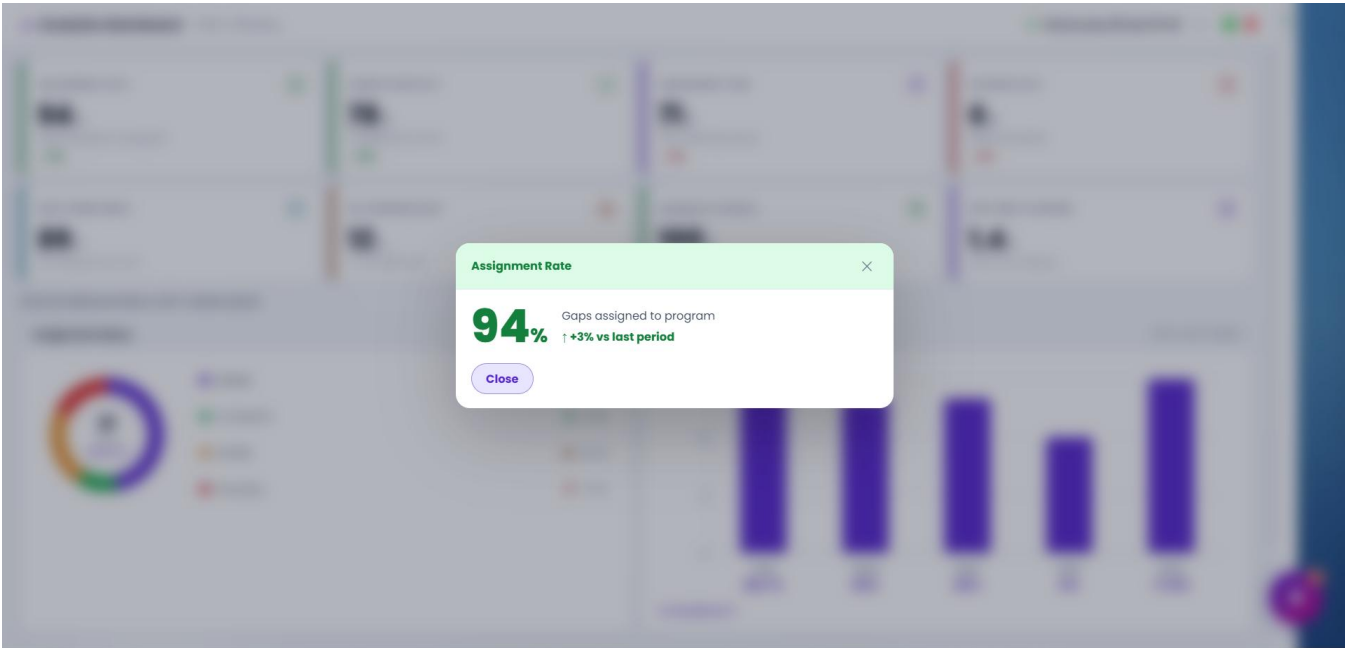
Section 16 — Final Response Object Example

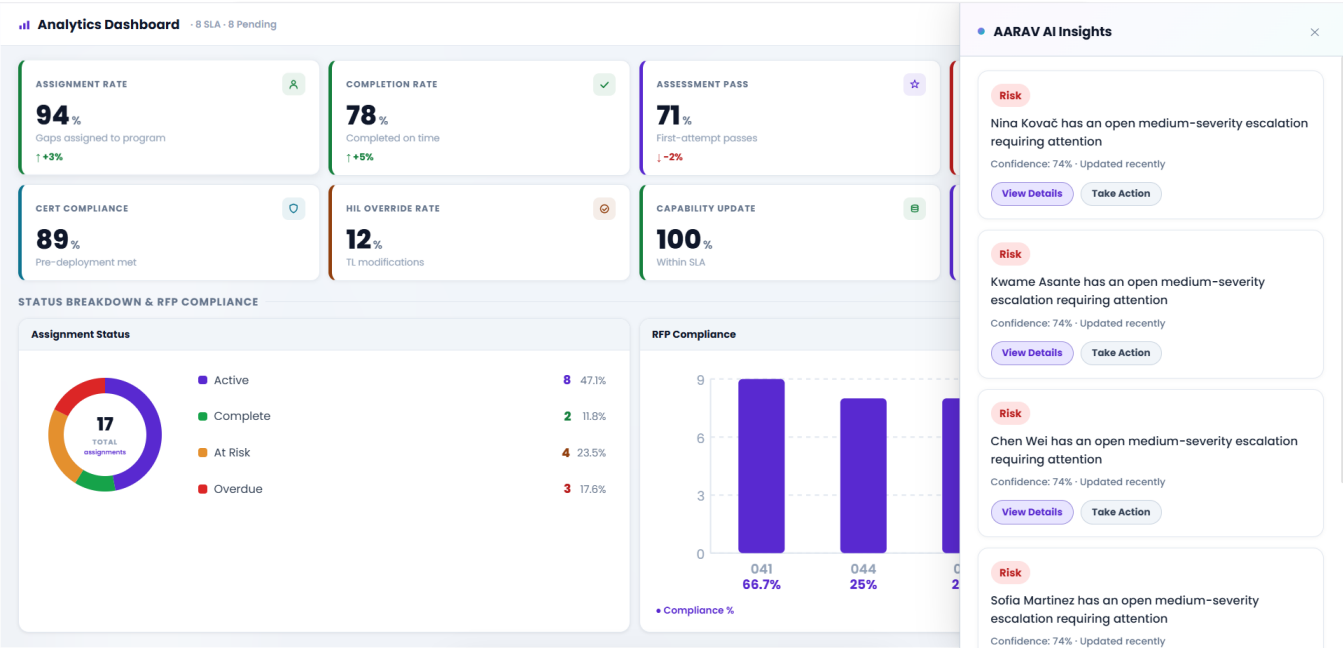
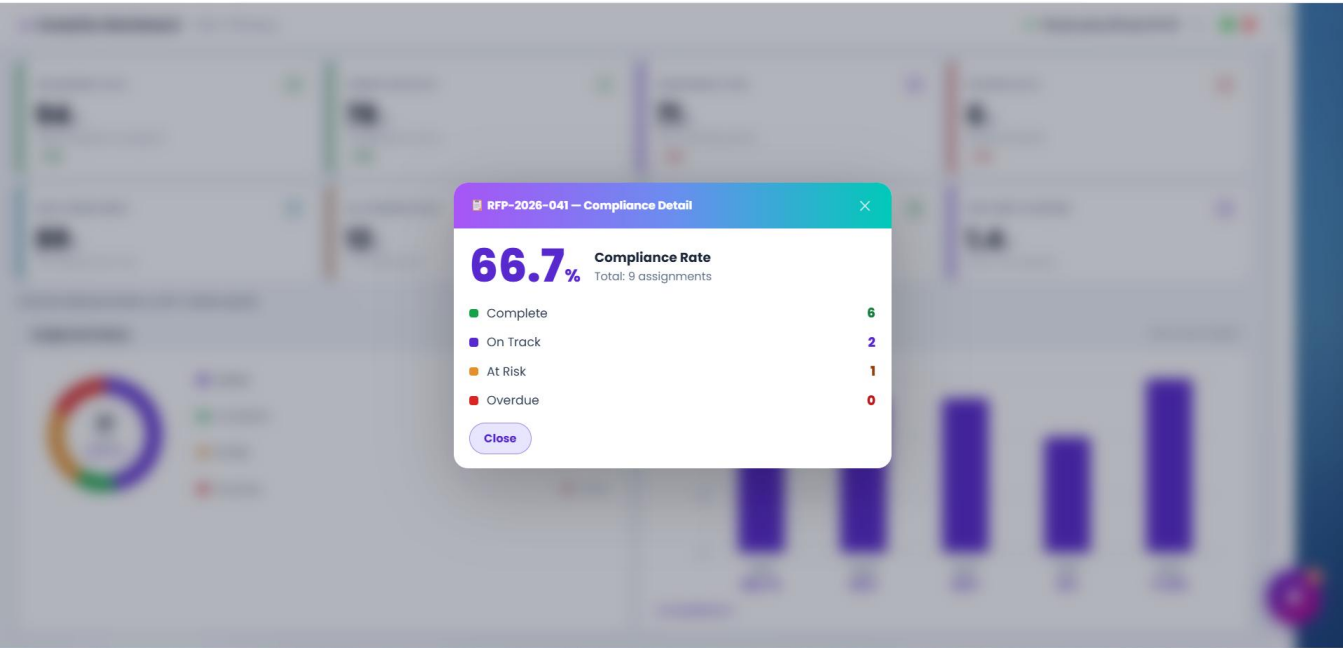
```
{  "flow id": "FLOW-AARAV-TN-FULL-DEMO",  "status": "completed",  "screens visited": [    "SCR-001-DESKTOP", "SCR-002-ANALYTICS", "SCR-003-TRACKER",    "SCR-005-HIL", "SCR-004-ESCALATIONS", "SCR-006-CERTS",    "SCR-007-AUDIT", "SCR-008-AARAV-ORB"  ],  "actions performed": [    { "action": "kpi_drilldown", "target": "kpi-assign-card", "result": "modal_opened" },    { "action": "hil_approve", "resource": "{name}", "result": "row_removed" },    { "action": "extend_deadline", "resource": "{name}", "new deadline": "{date}", "result": "escalation resolved" },    { "action": "cert journey view", "resource": "{name}", "result": "modal opened" }  ],  "humanoid_final_spoken_summary": "We opened the Talent Nurturing System, reviewed all 8 KPIs in Analytics, inspected active assignments in the Tracker, approved a pending HIL recommendation, extended a deadline in the Escalations panel, reviewed a certification journey, read the latest audit events, and checked the Aarav AI insight panel. All six modules are operational and connected to the live database.",  "recommended next action": "Seed additional assignment data and re-run demo for full coverage." }
```

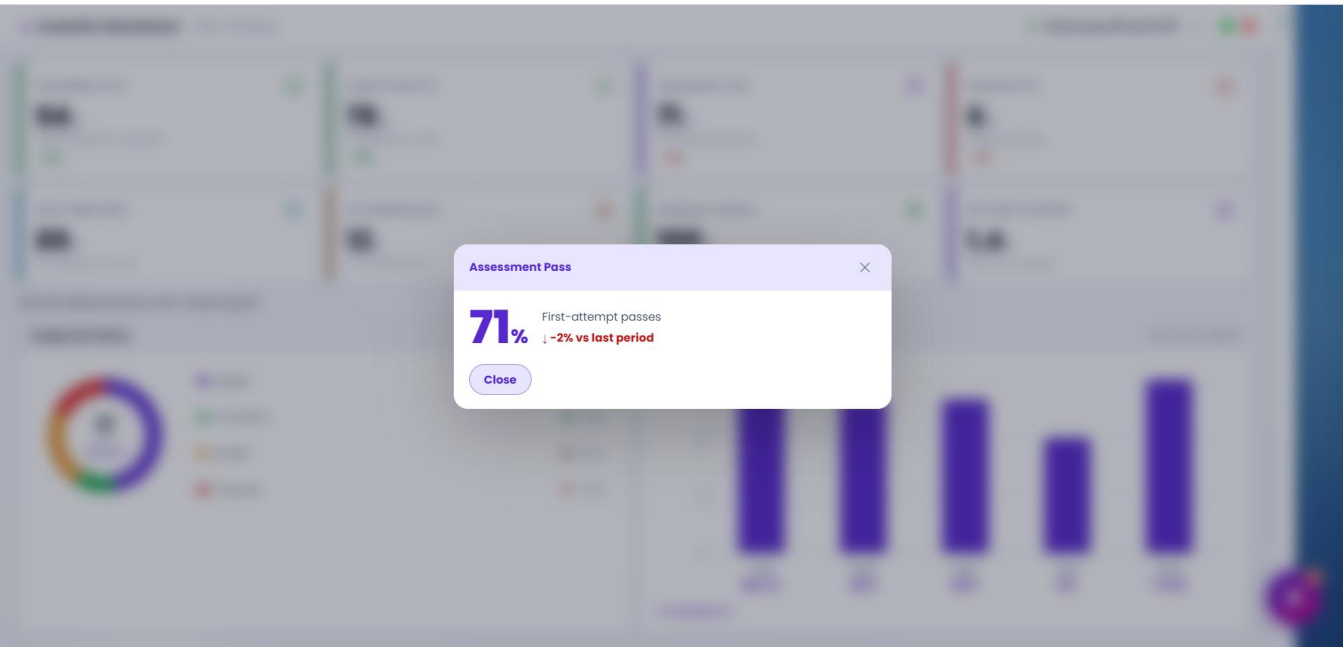
Screenshots / Evidence Images

Insert the corresponding application screenshots at each placeholder below. Screenshots should be captured at 1440×900 and tagged with the Screen ID (e.g. SCR-001-DESKTOP, SCR-002-ANALYTICS, etc.).









Assignment Tracker

- 8 SLA - 8 Pending

Wednesday 29 April 21:52

All	Active	At Risk	Overdue	Complete			
Resource ID	Name	Program	RFP	C - T - CS	Progress	Deadline	Status
R-1125	Daniel Osei Compliance Officer	Risk Management Fundamentals	RFP-2026-080		35%	2026-05-30	active
R-1133	Kwame Asante DevOps Engineer	DevOps Foundations	RFP-2026-086		20%	2026-04-30	at risk
R-1129	Nina Kovač Data Privacy Lead	Data Privacy Practitioner	RFP-2026-083		62%	2026-05-28	active
R-1108	Chen Wei Cybersecurity Lead	AI Ethics & Safety	RFP-2026-071		18%	2026-05-15	at risk
R-1104	James Okonkwo Cloud Architect	Cybersecurity Professional	RFP-2026-062		45%	2026-04-22	overdue
R-1112	Amara Diallo Product Manager	Agile Project Management	RFP-2026-065		50%	2026-05-25	active
R-1117	Ravi Patel AI Engineer	Data Governance Certification	RFP-2026-068		30%	2026-05-28	active
R-1121	Sofia Martinez Data Analyst	PMP Exam Prep	RFP-2026-074		15%	2026-04-26	overdue
Fatima Al-Bekid							
C = Content · T = Online Test · CS = Case Study							
17 of 17 records							

A

5

35%

In progress

Resource ID

R-1125

Full Name

Daniel Osei

Role

Compliance Officer

RFP

RFP-2026-080

Program

Risk Management Fundamentals

Deadline

2026-05-30

Content

In Progress

Online Test

Not Started

Case Study

Not Started

Attempts

Notes

Escalation Panel

8 SLA - 8 Pending

Wednesday 29 April 21:53

STATUS	RESOURCE	ID	PROGRAM	DUE	CONTENT	TEST	OVERALL	DAYS LEFT	ACTIONS
Resolved - replace	Daniel Osei	R-1125	Risk Management Fundamentals	2026-05-30	35%	0%	35%	31	
At Risk	Nina Kovač	R-1129	Data Privacy Practitioner	2026-05-28	100%	25%	62%	29	ExtendReplaceAccept
At Risk	Kwame Asante	R-1133	DevOps Foundations	2026-04-30	20%	0%	20%	1	ExtendReplaceAccept
At Risk	Chen Wei	R-1108	AI Ethics & Safety	2026-05-15	18%	0%	18%	16	ExtendReplaceAccept
Resolved - extend	Ravi Patel	R-1117	Data Governance Certification	2026-05-28	30%	0%	30%	29	
Resolved - accept risk	Amara Diallo	R-1112	Agile Project Management	2026-05-25	100%	50%	50%	26	
Overdue	Sofia Martinez	R-1121	PMP Exam Prep	2026-04-26	15%	0%	15%	-3	ExtendReplaceAccept
At Risk	Priya Sharma	R-1101	Cloud Security Fundamentals	2026-05-10	22%	0%	22%	10	ExtendReplaceAccept
Overdue	James Okonkwo	R-1104	Cybersecurity Professional	2026-04-22	100%	45%	45%	-6	ExtendReplaceAccept
At Risk	Yemi Adeyemi	R-1063	ISO 27001 Security Framework	2026-04-28	28%	0%	28%	5	ExtendReplaceAccept
Overdue	Li Wei	R-1077	GDPR Data Handling	2026-04-25	10%	0%	10%	-2	ExtendReplaceAccept

8 open - 11 total

A

HIL Approval Queue

- 8 SLA - 8 Pending

Wednesday 29 April 21:54

RESOURCE	ID	PROGRAM	RFP	MODULES	EST. HOURS	DEADLINE	STATUS	ACTIONS
Kwame Asante	R-1133	DevOps Foundations	RFP-2026-086	3	11h	2026-04-30	Pending HIL	Approve Reject
Nina Kovač	R-1129	Data Privacy Practitioner	RFP-2026-083	3	10h	2026-05-28	Pending HIL	Approve Reject
Amara Diallo	R-1112	Agile Project Management	RFP-2026-065	3	9h	2026-05-25	Pending HIL	Approve Reject
Ravi Patel	R-1117	Data Governance Certification	RFP-2026-068	3	9h	2026-05-28	Pending HIL	Approve Reject
Yemi Adeyemi	R-1063	Cybersecurity Professional	RFP-2026-062	3	13h	2026-05-28	Approved	—
Chen Wei	R-1108	AI Ethics & Safety	RFP-2026-071	3	11h	2026-05-15	Pending HIL	Approve Reject
Fatima Al-Rashid	R-1088	GDPR Data Handling	RFP-2026-051	3	9h	2026-05-20	Pending HIL	Approve Reject
Priya Sharma	R-1101	Cloud Security Fundamentals	RFP-2026-058	3	11h	2026-05-10	Pending HIL	Approve Reject
Yemi Adeyemi	R-1063	ISO 27001 Security Framework	RFP-2026-044	3	13h	2026-04-28	Pending HIL	Approve Reject
Marcus Chen	R-1051	NLP Evaluation Certification	RFP-2026-047	3	11h	2026-05-03	Approved	—

8 pending · 10 total

Certification Registry

- 8 SLA - 8 Pending

Wednesday 29 April 21:55

AllRegisteredVerifiedPending

CERTIFICATION	RESOURCE	ID	ROLE	VERIFIED	DEPLOYMENT	CAP. REGISTER	STATUS
CIPP/E Certificate	Nina Kovač	R-1129	Data Privacy Lead	2026-04-29	—	Pending	Verified
LF DevOps Certificate	Kwame Asante	R-1133	DevOps Engineer	—	—	Pending	Pending
PMI-ACP Certificate	Amara Diallo	R-1112	Product Manager	2026-04-30	—	Pending	Verified
AI Safety Certificate	Chen Wei	R-1108	Cybersecurity Lead	—	—	Pending	Pending
IRM Certificate	Daniel Osei	R-1125	Compliance Officer	—	—	Pending	Pending
PMP Certification	Sofia Martinez	R-1121	Data Analyst	—	—	Pending	Pending
CDMP Associate	Ravi Patel	R-1117	AI Engineer	—	—	Pending	Pending
NLP Eval Certificate	Marcus Chen	R-1051	NLP Specialist	2026-04-25	—	Pending	Verified
ML Fairness Certificate	Fatima Al-Rashid	R-1088	ML Engineer	2026-04-22	2026-04-29	Updated	✓ Registered
CISSP Associate	James Okonkwo	R-1104	Cloud Architect	2026-04-28	—	Pending	Verified
AWS Cloud Practitioner	Oliver Chen	R-1044	Data Scientist	—	—	Pending	Pending

Audit Log

- 8 SLA - 8 Pending

Wednesday 29 April

TIME	LEVEL	RFP	RESOURCE ID	ACTION	ACTOR	MESSAGE
29/04/2026, 21:01	warn	RFP-2026-086	R-1133	at_risk_detected	system	At-risk alert: Kwame Asante — 20% with 1 day re
29/04/2026, 21:01	action	RFP-2026-080	R-1125	escalation_resolved	talent_lead_01	Escalation resolved (replace): Daniel Osei
29/04/2026, 21:01	info	RFP-2026-086	R-1133	hil_pending	system	HIL review pending — Kwame Asante, DevOps
29/04/2026, 21:01	info	RFP-2026-083	R-1129	hil_pending	system	HIL review pending — Nina Kovač, Data Privacy
29/04/2026, 20:50	error	RFP-2026-062	R-1104	overdue_detected	system	Assignment overdue: James Okonkwo â€” Cyb deadline missed
29/04/2026, 20:50	warn	RFP-2026-058	R-1101	at_risk_detected	system	At-risk alert: Priya Sharma 22% with 10 days rem
29/04/2026, 20:50	error	RFP-2026-062	R-1104	escalation_raised	system	Escalation raised for James Okonkwo â€” over
29/04/2026, 20:50	warn	RFP-2026-	R-1108	at_risk_detected	system	At-risk alert: Chen Wei 18% with 16 days remaini

Audit Log

- 8 SLA - 8 Pending

Wednesday 29 April 21:55

TIME	LEVEL	RFP	RESOURCE ID	ACTION	ACTOR	MESSAGE
29/04/2026, 21:01	warn	RFP-2026-086	R-1133	at_risk_detected	system	At-risk alert: Kwame Asante — 20% with 1 day remaining
29/04/2026, 21:01	action	RFP-2026-080	R-1125	escalation_resolved	talent_lead_01	Escalation resolved (replace): Daniel Osei
29/04/2026, 21:01	info	RFP-2026-086	R-1133	hil_pending	system	HIL review pending — Kwame Asante, DevOps
29/04/2026, 21:01	info	RFP-2026-083	R-1129	hil_pending	system	HIL review pending — Nina Kovač, Data Privacy
29/04/2026, 20:50	error	RFP-2026-062	R-1104	overdue_detected	system	Assignment overdue: James Okonkwo â€” Cybersecurity deadline missed
29/04/2026, 20:50	warn	RFP-2026-058	R-1101	at_risk_detected	system	At-risk alert: Priya Sharma 22% with 10 days remaining
29/04/2026, 20:50	error	RFP-2026-062	R-1104	escalation_raised	system	Escalation raised for James Okonkwo â€” overdue
29/04/2026, 20:50	warn	RFP-2026-071	R-1108	at_risk_detected	system	At-risk alert: Chen Wei 18% with 16 days remaining
29/04/2026, 20:50	info	RFP-2026-065	R-1112	hil_pending	system	HIL review pending â€” Amara Diallo, Agile PM program
29/04/2026, 20:50	info	RFP-2026-	R-1117	hil_pending	system	HIL review pending â€” Paul Patel, Data Governance

[SCR-001-DESKTOP VIEW — insert screenshot here]

[SCR-002-ANALYTICS VIEW — insert screenshot here]

[SCR-003-TRACKER VIEW — insert screenshot here]

[SCR-004-ESCALATIONS VIEW — insert screenshot here]

[SCR-005-HIL VIEW — insert screenshot here]

[SCR-006-CERTS VIEW — insert screenshot here]

[SCR-007-AUDIT VIEW — insert screenshot here]

[SCR-008-AARAV-ORB VIEW — insert screenshot here]